

OpenVLA-OFT · arXiv 2502.19645 · Kim, Finn, Liang (Stanford)

Empiricist

11+3

Signature dump + $K \in \{1, 4, 8\}$ sweep
P0/P1/Defer · GPU honesty

Oct 26 paper club · VLA Architectures

SPINE: AR discrete bins → parallel continuous L1 chunks

OFT $\neq$ FAST $\neq$ FM · FT recipe / speed+success patch · not $\pi_{0.5}$ on BEHAVIOR

P0 · SIGNATURE DUMP

Assert shapes before any SR claim

Vanilla AR

out = 7 bin token ids

dtype: int / long

shape: (B, 7) · K=1

dequantize → EE delta

OFT Cont-L1

out = continuous tensor

dtype: float32 / bf16

shape: (B, K, D) · $R^{\{K \times 7\}}$

K=8 LIBERO default

Speed smoke (same GPU, 100 queries)

Expect $\geq 20\times$ actions/s (paper $26\times$ on A100) · log max_memory_allocated

Fail bar: same discrete head as vanilla · $< 5\times$ speedup · silent CPU “GPU run”

OUTPUT SIGNATURE SHIFT

Vanilla OpenVLA → OpenVLA-OFT

Axis	Vanilla OpenVLA	OFT
Decode	Serial AR discrete bins	Parallel continuous decode
Chunk	K=1	K=8 LIBERO / K=25 ALOHA
Head	256-bin CE	Continuous MLP + L1
Hz	~4.2	~109.7 LIBERO (26×); ALOHA ~77.9 (43×)
LIBERO SR	76.5%	97.1% (π ₀ 94.2%)

OFT+ adds FiLM on SigLIP+DINOv2 (multi-view language) — without FiLM, language → ~chance

K-SWEEP · Cont-L1 chunk size

Hypothesis: $K > 1$ raises Long SR + effective Hz; $K = 1 \approx$ vanilla latency regime

K=1

baseline

Near-vanilla
latency regime
no chunk benefit

K=4

scout

Mid chunk
partial Long gain
Hz ↑

K=8

primary

Paper LIBERO
default
~109.7 Hz

ALOHA uses $K=25$ @ 25 Hz — execute full chunk open-loop

Normalize closed-loop by $\text{commit_seconds} = K / \text{control_Hz}$ when comparing DP/ACT/ π_0

Log every run: (sig, K, D, Hz, latency_s, commit_s, VRAM, lora_r, steps, suite, SR)

EMPIRICIST · Priority matrix

P0 Y · P1 Partial · P2 · Defer

Pri	Experiment	Smoke	GPU-h
P0	Vanilla vs OFT Cont-L1 dry infer + signature dump	Y	<1
P1	LIBERO-Spatial LoRA; PD&AC disc vs Cont-L1; $K \in \{1,4,8\}$	Partial	16-60
P1	wrist+proprio; L1 vs Cont-Diff; vs π_0 /DP/ACT	Partial	20-80
P2	FiLM on/off; FAST AR vs OFT parallel	Partial	12-40
Defer	Paper 8×80GB full suites; real ALOHA	N	hundreds

Budget: P0 <1 GPU-h · LIBERO scout 16-40 · Env: /workspace/openvla-oft-quiet/env_outline/

EMPIRICIST HYPOTHESES

H-sig

$\geq 20\times$ actions/s Cont-L1 K=8 vs vanilla same GPU

H-sr

Spatial LoRA scout trend $\gg$ 76.5% vanilla

H-cont

Cont-L1 $\geq$ discrete PD&AC; $\approx$ Cont-Diff cheaper

H-chunk

K $\uparrow$ helps Long/smoothness more than Spatial

H-io

wrist+proprio $\uparrow$ SR; Hz still $\gg$ vanilla

H-film

multi-view without FiLM $\rightarrow$ chance; LIBERO weak

EXACT OFT RECIPE

Sec V-E / App. A — six ingredients (OFT+ = #6)

1

Parallel decoding

Bidirectional attn over
empty action embeds

2

Action chunking

K=8 LIBERO /
K=25 ALOHA

3

Continuous MLP head

4-layer ReLU;
drop 256-bin LM head

4

L1 on [-1,1]

Mean L1 regression
(train until L1 < 0.01)

5

LoRA r=32

Optional wrist +
proprio projector

6

OFT+: FiLM

Lang $\rightarrow \gamma, \beta$ on SigLIP
+ DINOv2 blocks

BOARD TABLE · Hz / SR

Memorize these cells — Stakeholder tattoo

LIBERO

SR avg	76.5% → 97.1%	+20.6 pp
Hz (A100)	4.2 → 109.7	26×
vs π_0 SR	π_0 94.2%	OFT wins
+wrist+proprio Hz	71.4	still $\gg$ vanilla

ALOHA (OFT+)

Hz	1.8 → 77.9	43×
π_0 Hz still	291.6	JAX faster
Scoop	100 vs π_0 93.3	OFT+ best
Put-in-pot	51 vs π_0 42	~half; hard

Latency (tables)

LIBERO Cont-L1 ~ 0.073 s
ALOHA OFT+ 0.321 s
(π_0 0.086 s)

Abstract caveat

“0.07 ms / 0.321 ms”
conflicts with Table II/III
Prefer seconds

Trainable

LIBERO OFT 279M
ALOHA OFT+ 853M
(FiLM +456M)

Rule: OpenVLA/HF/LoRA ops + high Hz → OFT mandatory; stock AR obsolete for 25-50 Hz

Max wall-clock + OpenPI/JAX → $\pi_0/\pi_{0.5}$ still win latency + pretrain coverage

COST / GPU / DEMO BUDGET

FT GPUs

8× A100/H100 80GB · DDP + LoRA

LIBERO wall

1-2 days · 50-150K steps / suite

ALOHA demos

fold 20-30 · scoop 45 · pot 300

Trainable

279M LIBERO · 853M OFT+ (FiLM)

Scout budget

P0 <1 GPU-h · Partial 16-40

Infer

Single A100; 7B still heavier than ACT/DP/ π_0

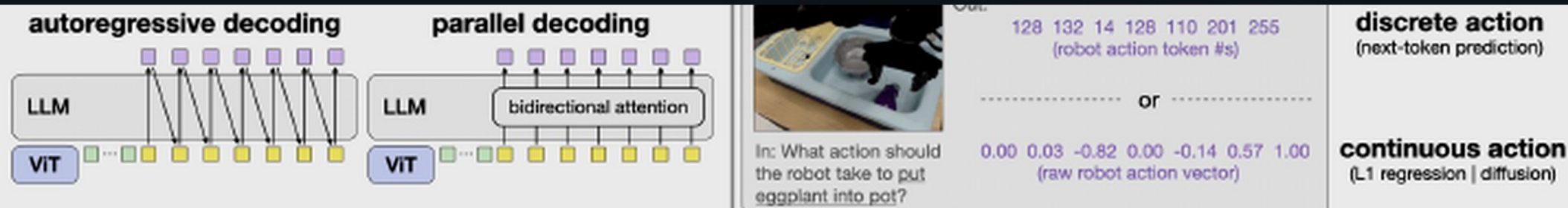


Fig. 2: Key design decisions for VLA fine-tuning. Left: Comparison between autoregressive decoding, which generates actions sequentially, and parallel decoding, which leverages bidirectional attention and generates all actions in a single forward pass. Right: Comparison between discrete action tokens with next-token prediction and continuous action values with L1 regression or diffusion modeling objectives. The original OpenVLA training scheme includes autoregressive decoding, discrete actions, and next-token prediction.

ing. While these diffusion-based VLAs achieve higher action throughput than autoregressive VLAs by generating multi-timestep action chunks simultaneously, they introduce computational trade-offs through slower training and multiple denoising or integration steps at inference time. Furthermore, these diffusion VLAs vary considerably in architecture, learning algorithm, vision-language fusion approach, and input-output specifications—and which design elements most significantly impact performance remains unclear. Through controlled experiments, we show that policies fine-tuned with a simpler L1 regression objective can match more complex approaches in

timestep action takes 0.33 seconds on an NVIDIA A100 GPU. For a chunk size of K timesteps and action dimensionality D , OpenVLA requires KD sequential decoder forward passes versus just D passes without chunking. This K -fold increase in latency makes action chunking impractical for high-frequency robots under the original formulation. In the next section, we present a parallel generation scheme that enables efficient action chunking.

IV. STUDYING KEY VLA FINE-TUNING DESIGN DECISIONS

Fig.3 · LIBERO four suites

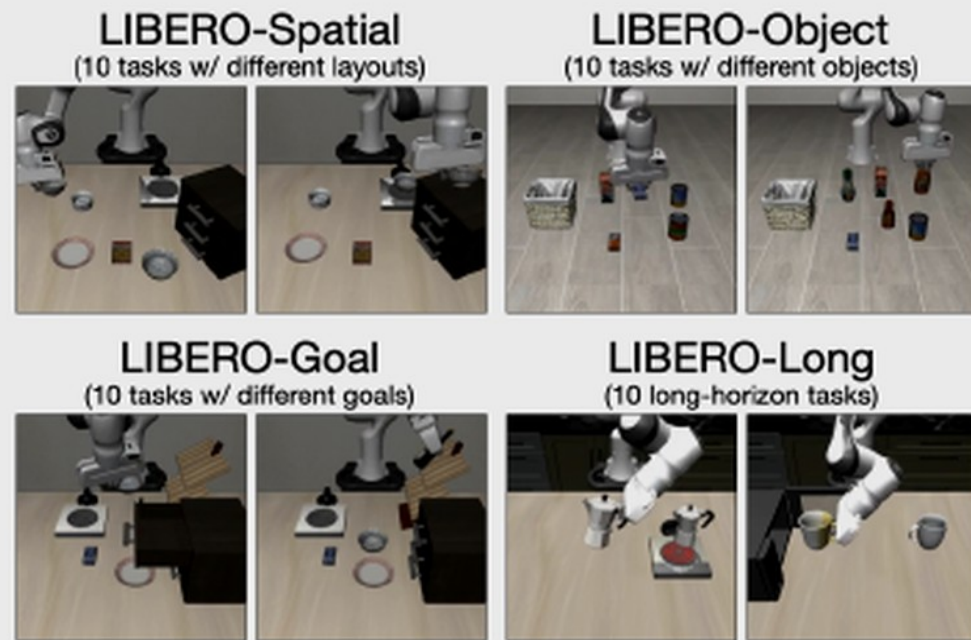


Fig. 3: LIBERO simulation benchmark [26] task suites. We study VLA fine-tuning design decisions using four representative task suites. Here we depict two of ten tasks per suite.

embeddings so that γ and β influence all patch embeddings. Concretely, this makes γ and β D_{ViT} -dimensional vectors, where D_{ViT} is the number of hidden dimensions (i.e., the number of elements in each of the patch embeddings in the vision transformer's latent representations).

We apply FiLM after the self-attention layer and before the feedforward layer in each vision transformer block, with separate projectors for each block (see Figure 8). Additional

for non-diffusion methods and 100-250K steps for diffusion methods (which converge slower), using a batch size of 64-128 across 8 A100/H100 GPUs. We test checkpoints every 50K steps and report the best performance for each run. Unless specified otherwise, policies receive one third-person image and language instruction as input. For methods using action chunking, we set chunk size to $K = 8$ to match the Diffusion Policy baseline [5], and execute full chunks before replanning, which we find improves both speed and performance. See Appendix D for hyperparameter details.

Our primary baseline in this study is the base OpenVLA model fine-tuned using the original fine-tuning recipe. However, for broader comparison, we also include LIBERO results from prior state-of-the-art imitation learning methods, such as Diffusion Policy [5], Octo [49], DiT Policy [13], Seer [50], MDT [40], and π_0 [3]. Note that Seer uses additional LIBERO-90 pretraining data.

B. LIBERO Task Performance Comparisons

For satisfactory deployment, robot policies must demonstrate reliable task execution. We first assess how different VLA fine-tuning design decisions affect success rates on the LIBERO benchmark.

Our efficiency analysis (which we discuss later) reveals that parallel decoding (PD) and action chunking (AC) together are necessary for high-frequency control (25-50+ Hz), especially for bimanual robots with double the amount of action

Empiricist · close

P0: signature dump + Cont-L1 vs vanilla Hz. P1: K sweep + LIBERO scout.

Budget P0 <1 GPU-h · scout 16-40 · Defer 8×80GB / real ALOHA

Q&A · Oct 26 · OpenVLA-OFT / 2502.19645

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